step1_data_preparation

November 8, 2025

[5]:
```python
#Import libraries

import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import f1_score, accuracy_score, classification_report

print("Libraries loaded.")
```

```
Libraries loaded.
```

[6]:
```python
#Load datasets and preview

train = pd.read_csv("../../Dataset/high_salary.csv")
live = pd.read_csv("../../Dataset/high_salary.live.csv")

print("Train shape:", train.shape)
print("Live shape:", live.shape)
```

```
Train shape: (20900, 19)
Live shape: (6967, 18)
```

[7]:
```python
display(train.head())
```

```
      id  social-security-number  house-number  age-group     workclass  \
0   8616             552701574.0        9854.0        2.0  self-emp-inc
1  21982             956556990.0        7588.0        3.0     local-gov
2  11191             958358623.0        6729.0        0.0       private
3  22229             224206693.0        6288.0        2.0       private
4  20732             276413230.0        8276.0        3.0       private

     fnlwgt  education  education-num      marital-status         occupation  \
```

```
0  270079.0  bachelors          13.0  married-civ-spouse    exec-managerial
1  146325.0  doctorate          16.0  married-civ-spouse     prof-specialty
2  240767.0    hs-grad           9.0       never-married      other-service
3  118536.0    hs-grad           9.0            divorced  machine-op-inspct
4  160440.0  bachelors          13.0  married-civ-spouse              sales

     relationship   race     sex  capitalgain  capitalloss  hoursperweek  \
0         husband  white    male          0.0          0.0           3.0
1         husband  white    male          0.0          2.0           2.0
2   not-in-family  white  female          0.0          0.0           1.0
3  other-relative  black    male          0.0          0.0           2.0
4         husband  white    male          0.0          0.0           3.0

  native-country-code native-country  label
0                 USA  united-states    1.0
1                 USA  united-states    1.0
2                 USA  united-states    0.0
3                 USA  united-states    0.0
4                 USA  united-states    1.0
```

[8]:
```
#Save temporary files for next steps

train.to_csv("Load_model/train_temp.csv", index=False)
live.to_csv("Load_model/live_temp.csv", index=False)

print("Saved train_temp.csv and live_temp.csv for next steps.")
```

```
Saved train_temp.csv and live_temp.csv for next steps.
```